University of Hertfordshire

School of Physics, Engineering and Computer Science

Module code: 7COM1085-0509-2025 – Research Methods

The Impact of Feature Selection on the Performance of Malware Detection Systems

Final Report

Group No 8

Esika Arifin Rumky – 23027070

MD Imran Ahmed – 23032788

Dhanyashree Madappa Nethravathi – 23085256

Md Tanvir Islam – 23027714

ASM Mahmudur Rahman – 23074900

Table of Contents

Table of Contents 2
1. Introduction 3
2. Background 3
3. Literature Review Protocol 4
3.1 Research Question and PICO Elements 4
3.2 Search Strategy 5
3.3 Inclusion and Exclusion Criteria 5
4. Literature Review 5
5. Conclusion 7
Bibliography 8

1. Introduction

Malware continues to spread rapidly and remains a major cybersecurity problem, with over 560,000 new samples identified every day and global cybercrime costs estimated at around $8 trillion in 2023 (Abualhaj et al., 2026). Machine learning-based detection methods have become a leading alternative to signature-based approaches because they can identify previously unseen threats through pattern recognition rather than exact matching (Raza et al., 2026). However, these models carry a significant drawback: the large number of features required for accurate detection creates substantial computational overhead, which limits real-time deployment and scalability (Choudhary, Pawar and Haribhakta, 2025). Feature selection techniques have been proposed to address this problem, aiming to reduce the number of features used by a model while maintaining, or even improving, detection accuracy (Ele et al., 2025).

This final report presents a systematic literature review examining how effective feature selection is within malware detection systems. The review investigates whether feature selection, compared with using the full set of extracted features, improves both detection accuracy and computational efficiency. Building on the protocol established in the Interim Report, this report incorporates formative feedback received, revises the Introduction, Background and Research Protocol chapters, and adds the completed Literature Review (Chapter 4) and Conclusion (Chapter 5). Chapter 3 restates the research question and PICO framework, the finalised IEEE Xplore search string, and the inclusion and exclusion criteria used to screen candidate studies. Chapter 4 presents the data extraction and synthesis of the four studies that passed the review protocol, and Chapter 5 concludes by answering the research question using the evidence gathered.

2. Background

Malware detection has evolved considerably since the emergence of computer viruses in the 1970s (Choudhary, Pawar and Haribhakta, 2025). Signature-based detection was historically the dominant approach, matching files against known malicious patterns; however, this method has become increasingly ineffective against modern attacks that rely on obfuscation, polymorphism and zero-day exploits (Maganur et al., 2025). Consequently, machine learning methods have become widely adopted, using features extracted from binaries, network traffic or application behaviour to identify suspicious patterns (Kharnotia, Arora and Kour, 2026).

A persistent challenge is that feature extraction typically produces very high-dimensional vectors, creating a substantial computational burden. For example, the widely used EMBER vectorizer generates 2,381-dimensional feature vectors from Windows portable executables (Choudhary, Pawar and Haribhakta, 2025). Processing such high-dimensional data demands considerable memory, extends training time, and increases inference latency, which makes real-time deployment difficult (Choudhary, Pawar and Haribhakta, 2025; Khan et al., 2025).

Recent research has explored a range of feature selection strategies to address these challenges. Bio-inspired optimisation frameworks that combine Harris Hawks Optimisation with the Bat Algorithm have achieved detection accuracies of up to 99.52% while substantially reducing feature dimensionality (Abualhaj et al., 2026). Similarly, reinforcement learning-based sequential feature selection has achieved 99.22% accuracy using only 61 of 1,795 available features, a 96.6% reduction

that yields a 30.1-times improvement in computational efficiency over traditional ensemble methods (Khan et al., 2025). Hybrid filter-wrapper approaches applied to network traffic features have likewise achieved high classification accuracy while reducing the size of the feature space (Ele et al., 2025).

These findings suggest that feature selection is a viable means of balancing accuracy with efficiency in malware detection. However, the literature remains fragmented: studies use different datasets, feature types and evaluation metrics that are not always directly comparable, and several existing reviews focus on describing detection techniques rather than systematically evaluating the impact of feature selection itself (Raza et al., 2026; Kharnotia, Arora and Kour, 2026). A systematic review is therefore necessary to consolidate evidence on whether feature selection consistently improves detection accuracy and computational efficiency across different malware detection contexts.

3. Literature Review Protocol

3.1 Research Question and PICO Elements

Table 1 presents the research question in terms of the Population, Intervention, Comparison and Outcome (PICO) framework. The Population is malware datasets containing both malicious and benign software samples used for machine learning classification. The Intervention is the application of a feature selection technique, whether a filter, wrapper, embedded or hybrid method, applied to the dataset prior to model training. The Comparison contrasts this against a baseline that uses the complete set of extracted features. The Outcome combines two measurable dimensions: detection accuracy, captured through metrics such as precision, recall, F1-score and AUC, and computational efficiency, captured through training time, inference time, memory usage and the degree of dimensionality reduction achieved. Framing the outcome around both dimensions ensures the review can assess whether feature selection represents a genuine improvement rather than a trade-off that sacrifices one property for the other.

Table 1: PROTOCOL Part 1 – Research Question with PICO Elements

Elements	**Content**
Topic	The impact of feature selection on the performance of malware detection systems.
Research Question	Among malware datasets, does feature selection, compared with using all extracted features, improve malware detection accuracy and computational efficiency?
Context	The field of cybersecurity, specifically focusing on malware detection and the optimisation of machine learning models used for this purpose.
Population	Malware datasets consisting of malicious and benign software samples used for machine learning classification.
Intervention	The application of feature selection techniques (e.g., filter methods, wrapper methods, embedded methods, or hybrid approaches) to the dataset before model training.
Comparison	Using all extracted features (the baseline approach) versus using a selected subset of features.
Outcome	A systematic comparison measuring (1) detection accuracy (e.g., precision, recall, F1-score, AUC), and (2) computational efficiency (e.g., training time, inference time,

	memory usage, dimensionality reduction) between the two approaches.

3.2 Search Strategy

Table 2 shows the search string used to query the IEEE Xplore Digital Library. No changes to the search string were required following feedback on the Interim Report, as the string was confirmed to be syntactically correct and to return a manageable and relevant set of candidate papers.

Table 2: PROTOCOL Part 2 – Search String and Paper Count

Elements	Content
Search string	("feature selection" OR "feature reduction" OR "dimensionality reduction") AND ("malware detection" OR "malware classification") AND ("accuracy" OR "performance" OR "efficiency" OR "computational cost")
No. of papers	553 (Interim Report search). [TO UPDATE: re-run on IEEE Xplore before submission and record the current count and date here; export the results to papers.csv]

3.3 Inclusion and Exclusion Criteria

Table 3 lists the inclusion and exclusion criteria applied to the title and abstract of each returned paper. The inclusion criteria restrict the review to peer-reviewed, English-language, empirical studies published from 2018 onward that directly compare detection performance with and without feature selection, and that report both accuracy and efficiency metrics. The exclusion criteria remove studies outside the malware detection domain, papers that propose new architectures without evaluating feature selection, duplicated datasets, review or survey articles, and studies lacking quantitative results. Applying both sets of criteria together ensures that only primary, comparable, empirical evidence contributes to the data extraction and synthesis presented in Chapter 4.

Table 3: PROTOCOL Part 2 – Inclusion/Exclusion Criteria

Elements	Content
Inclusion Criteria	1. Studies that empirically compare malware detection performance with and without feature selection. 2. Studies that evaluate both detection accuracy (precision, recall, F1-score) and computational efficiency (training time, inference time, memory usage). 3. Studies published in peer-reviewed journals or conference proceedings. 4. Studies written in English. 5. Studies published from 2018 to the present to ensure contemporary relevance.
Exclusion Criteria	1. Studies focusing on detection methods not applicable to malware detection (e.g., intrusion detection, spam filtering). 2. Studies that only propose new detection architectures without evaluating feature selection impact. 3. Duplicated studies where the same data appears in multiple publications. 4. Review articles, book chapters, editorials, or incomplete papers (fewer than 4 pages). 5. Studies that do not report quantitative performance metrics.

4. Literature Review

Following the search and screening process described in Chapter 3, the exported IEEE Xplore results were screened against the inclusion and exclusion criteria using the five tracking columns in

papers.csv (include title, include abstract, exclude, accept contents, rq answers). Of the seven papers examined in detail, three were excluded because they are review or survey articles rather than primary empirical studies that measure detection performance with and without feature selection (Raza et al., 2026; Kharnotia, Arora and Kour, 2026; Maganur et al., 2025). The remaining four studies passed the full screening process and form the basis of the data extraction and synthesis presented in Tables 4 and 5.

Table 4 summarises the four accepted papers. Choudhary, Pawar and Haribhakta (2025) apply XGBoost-based feature selection and PCA to reduce EMBER's 2,381-dimensional malware feature vector to as few as 128 features. Ele et al. (2025) combine filter and wrapper feature selection methods on a 215-attribute Android network traffic dataset. Khan et al. (2025) use a Dueling Double Deep Q-Network to learn an adaptive, per-sample feature selection policy across two large malware datasets. Abualhaj et al. (2026) combine Harris Hawks Optimisation and the Bat Algorithm using both union and intersection selection strategies on a 79-feature URL-based malware dataset.

Table 4: Data Extraction Table

Paper	Feature Selection Method	Dataset	Key Result (RQ Answer, Cited Quote)
Choudhary, Pawar and Haribhakta (2025)	XGBoost-based selection and PCA	EMBER-2018, ERMDS, BODMAS (2,381 features)	"LightGBM trained on the 384-dimensional feature set after XGBoost feature selection achieves the highest accuracy of 97.52% on the unified dataset, providing an optimal balance between computational efficiency and detection performance" (Choudhary, Pawar and Haribhakta, 2025).
Ele et al. (2025)	Hybrid filter–wrapper selection	Android network traffic dataset (215 attributes)	"Among the classifiers, KNN remarkably achieved perfect scores with an accuracy of 100%, ROC-AUC of 1.000, precision of 1.000, and F1-score of 1.000" after the hybrid feature selection was applied (Ele et al., 2025).
Khan et al. (2025)	D3QN reinforcement-learning sequential selection	Microsoft Big2015 (1,795 features), BODMAS (2,381 features)	"D3QN achieves 99.22% and 98.83% accuracy while utilizing only 61 and 56 features on average... yield[ing] computational efficiency improvements of 30.1x and 42.5x over traditional ensemble methods" (Khan et al., 2025).
Abualhaj et al. (2026)	Harris Hawks Optimisation + Bat Algorithm (union/intersection)	ISCX-URL2016 (79 features)	"The HHO-BA (union) method demonstrated the best trade-off between computational cost and detection performance... Compared to the NoFs configuration, HHO-BA (union) reduced training time by 13.7% (XGBoost)... while achieving higher accuracy (99.52% vs. 99.48%)" (Abualhaj et al., 2026).

Three cross-cutting themes emerge from the data extraction table, each supported by evidence from at least two of the four accepted studies, as shown in Table 5.

The first theme, dimensionality reduction without accuracy loss, is the most consistent finding across the review. All four studies report that removing the majority of extracted features did not reduce, and in most cases slightly improved, detection accuracy (Choudhary, Pawar and Haribhakta, 2025;

Ele et al., 2025; Khan et al., 2025; Abualhaj et al., 2026). This directly answers the accuracy element of the research question: feature selection is not simply a compression technique but can act as a form of noise removal that helps classifiers generalise.

The second theme, computational efficiency gains, addresses the efficiency element of the research question. Three of the four studies report explicit reductions in training or inference time, ranging from a 13.7% training-time reduction using bio-inspired hybrid feature selection (Abualhaj et al., 2026) to a 30.1-times inference efficiency improvement using reinforcement-learning-based sequential feature selection (Khan et al., 2025). Choudhary, Pawar and Haribhakta (2025) similarly report that a reduced 384-feature model trained in 61 minutes generalised effectively to entirely unseen datasets, indicating that efficiency gains are not achieved at the expense of robustness.

The third theme, the accuracy–efficiency trade-off between selection strategies, shows that not all feature selection strategies behave identically. Abualhaj et al. (2026) found that a broader, union-based feature selection strategy produced the highest accuracy, while a stricter, intersection-based strategy produced the fastest training and inference times at a small accuracy cost. This suggests that the optimal feature selection strategy depends on the deployment context, favouring maximal accuracy in high-security applications and maximal speed in latency-sensitive or resource-constrained environments.

Table 5: Data Synthesis Table

Theme	Supporting Papers	Summary
Dimensionality reduction without accuracy loss	Choudhary, Pawar and Haribhakta (2025); Ele et al. (2025); Khan et al. (2025); Abualhaj et al. (2026)	All four studies removed the majority of extracted features (35%–97.6% depending on method) without reducing, and typically slightly improving, detection accuracy relative to the full-feature baseline.
Computational efficiency gains	Choudhary, Pawar and Haribhakta (2025); Khan et al. (2025); Abualhaj et al. (2026)	Three studies report explicit reductions in training or inference time, ranging from a 13.7% training-time reduction to a 30.1-times inference efficiency improvement, alongside lower memory and disk usage.
Accuracy–efficiency trade-off between selection strategies	Abualhaj et al. (2026); Khan et al. (2025)	Broader (union-based) feature subsets maximise accuracy, while stricter (intersection-based) or more aggressively reduced subsets maximise speed, indicating the optimal strategy depends on deployment context.

5. Conclusion

This report set out to answer the research question: among malware datasets, does feature selection, compared with using all extracted features, improve malware detection accuracy and computational efficiency? A systematic search of the IEEE Xplore Digital Library followed by inclusion and exclusion screening identified four primary empirical studies that directly compared feature-selected and full-feature approaches to malware detection.

The evidence gathered in Chapter 4 provides a clear answer to the research question. Across all four studies, feature selection reduced feature dimensionality substantially, by between 35% and 97.6%, without harming detection accuracy, and in most cases producing a small accuracy improvement over the full-feature baseline (Choudhary, Pawar and Haribhakta, 2025; Ele et al., 2025; Khan et al., 2025; Abualhaj et al., 2026). At the same time, three of the four studies demonstrated measurable computational efficiency gains, including reduced training time, reduced inference time, and lower memory and disk usage. The review also found that the specific choice of feature selection strategy, for example a union versus an intersection of candidate feature subsets, allows detection systems to be tuned toward either maximum accuracy or maximum speed depending on deployment requirements (Abualhaj et al., 2026).

Taken together, this evidence supports the conclusion that feature selection is an effective and largely reliable approach for improving both the accuracy and the computational efficiency of machine-learning-based malware detection systems, addressing the practical scalability concerns identified in Chapter 2. A limitation of this review is the small number of studies that met the full inclusion criteria, which reflects how much of the surrounding literature consists of survey and review papers rather than primary comparative studies. Future systematic reviews would benefit from a broader search across additional digital libraries and an expanded time window to strengthen the generalisability of these findings.

Bibliography

Abualhaj, M.M., Al-Mimi, H., Al-Zyoud, M., Al-Khatib, S.N., Daoud, M.Sh., Al-Aqrabi, H., Anbar, M. and Shalaldeh, A. (2026) 'A bio inspired hybrid optimization framework for efficient real time malware detection', Scientific Reports, 16(1). doi: 10.1038/s41598-025-33439-z.

Choudhary, A., Pawar, S. and Haribhakta, Y. (2025) Efficient Malware Detection with Optimized Learning on High-Dimensional Features. [online] arXiv.org. Available at: https://arxiv.org/abs/2506.17309.

Ele, S., Ele, B.I., Julius, M.S., Egete, D.O., Ele, M.S. and Anayo, O.H. (2025) 'Optimized Hybrid Feature Selection Model for Automatic Malware Detection and Classification Using Network Traffic Android Data', NIPES - Journal of Science and Technology Research, 7(3), pp. 127–139. doi: 10.37933/nipes/7.3.2025.09.

Khan, N., Al-Tamimi, A.Y., Bermak, A. and Khalil, I.M. (2025) Adaptive Malware Detection using Sequential Feature Selection: A Dueling Double Deep Q-Network (D3QN) Framework for Intelligent Classification. [online] arXiv.org. Available at: https://arxiv.org/abs/2507.04372 (Accessed: 30 June 2026).

Kharnotia, S., Arora, B. and Kour, R. (2026) 'Feature-driven static analysis for learning-based android malware detection: A review', ICT Express, 12(1), pp. 186–208. doi: 10.1016/j.icte.2026.01.005.

Maganur, S., Jiang, Y., Huang, J. and Zhong, F. (2025) 'Feature-Centric Approaches to Android Malware Analysis: A Survey', Computers, 14(11), p. 482. doi: 10.3390/computers14110482.

Raza, M.M., Umair, M., Choudhry, I.A., Qasim, M., Naseem, M.T., Asghar, M.N., Gavilanes, D., Vergara, M.M. and Ashraf, I. (2026) 'Malware Detection and AI Integration: A Systematic Review of Current Trends and Future Directions', Computer Modeling in Engineering & Sciences, 146(3), pp. 1–10. doi: 10.32604/cmes.2025.074164.